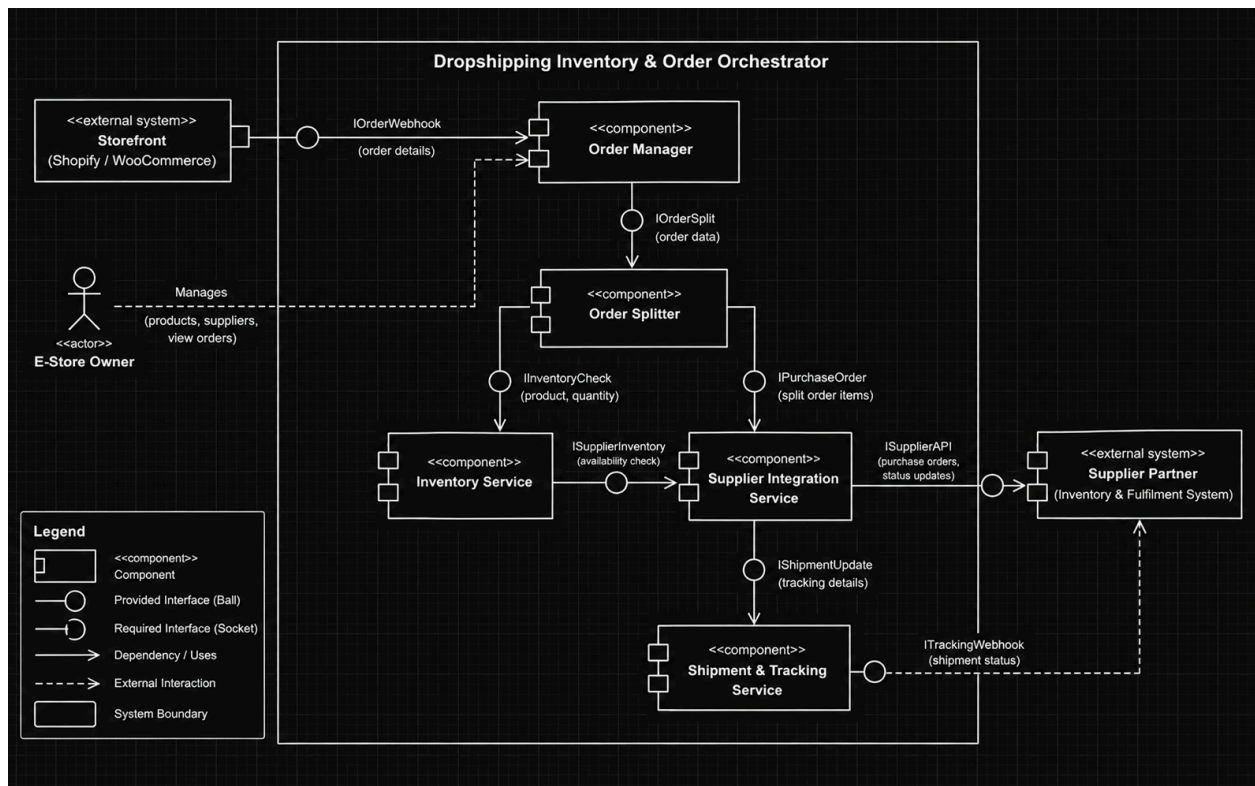


SOFTWARE ENGINEERING -03

Requirements Table

Dropshipping Inventory & Order Orchestrator

DELIVERABLE 1: UML COMPONENT DIAGRAM



DELIVERABLE 2: JUSTIFICATION

Architecture Selection: Microservices Architecture

We chose Microservices Architecture for the Dropshipping Inventory & Order Orchestrator.

Architectural Choice:

The system is divided into independent components such as Order Manager, Order Splitter, Inventory Service, Supplier Integration Service, and Shipment & Tracking Service. A microservices architecture is suitable because each service handles a specific responsibility and can communicate with other services through defined interfaces.

Reason 1:

The system needs to process orders from Shopify/WooCommerce, split multi-vendor orders, verify supplier inventory, dispatch purchase orders, and track shipments. Separating these functions into individual services provides clear separation of responsibilities and makes the system easier to maintain.

Reason 2:

Different Supplier Partners may have different inventory and order-processing APIs. The **Supplier Integration Service** can handle communication with suppliers independently, allowing supplier-specific changes without affecting the other services.

Security Advantage:

Each service can use authenticated and authorized interfaces for communication. This helps protect customer delivery addresses, order information, and supplier API connections from unauthorized access.

Performance Benefit:

Individual services can be independently scaled when the order volume increases. For example, the Order Manager and Inventory Service can handle increased webhook and inventory-check traffic without requiring the entire system to be scaled.